

Energy

(Scheduled Event - 30 Minutes)



Description: Participants will be assessed on their understanding of energy forms, energy transformations, energy transfer, and major renewable and nonrenewable energy sources, including real-world energy production systems.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Teams must bring: Pen/Pencil

Teams may not bring:

- Notes or guides of any kind.

Optional - teams may bring:

- Four-function calculator

Safety Requirements:

- None

The Competition: This event may be administered as a written test or as a series of stations which may include diagrams, models, data analysis, case studies of energy facilities, and problems for students to solve.

Content topics will include but are not limited to:

1. Energy Concepts

- Define energy and identify common forms including kinetic, potential, thermal, chemical, electrical, and nuclear energy.
- Apply the law of conservation of energy.
- Describe energy transformations in real-world systems.

2. Energy Transfer

- Describe conduction, convection, and radiation with real-world examples.
- Explain how energy is transferred and transformed in physical systems.

3. Renewable and Nonrenewable Energy Sources

- Identify renewable sources: solar, wind, hydroelectric, geothermal, and biomass.
- Identify nonrenewable sources: coal, oil, natural gas, and nuclear energy.
- Compare advantages, limitations, and environmental impacts of energy sources.

4. Representative Energy Case Studies (U.S. + Global)

Students should recognize each facility's energy type, general method of power generation, and scale/impact.

Bhadla Solar Park (India)	Solar Star (California, USA)
Hornsea Offshore Farm (United Kingdom)	Horse Hollow Energy Center (Texas, USA)

Three Gorges (China)	Hoover (Nevada/Arizona, USA)
Hellisheiði Power Plant (Iceland)	The Geysers Complex (California, USA)
Drax Power Station (United Kingdom)	Portland Biomass Energy Facility (Oregon, USA)
Barakah Power Plant (United Arab Emirates)	Palo Verde Generating Station (Arizona, USA)
Tuoketuo Power Station (China)	Gavin Power Plant (Ohio, USA)
Ras Tanura (Saudi Arabia)	Port Arthur Refinery (Texas, USA)
Surgut-2 Power Station (Russia)	West County Energy Center (Florida, USA)

5. Energy Use, Efficiency, and Environmental Impact

- Explain energy efficiency and energy loss in systems.
- Describe environmental impacts of energy production.
- Evaluate tradeoffs between energy sources (cost, reliability, sustainability).

6. Data Analysis and Applications

- Interpret diagrams, charts, graphs, tables, and case study information.
- Apply energy concepts to analyze systems and solve problems.

Examples:

- Interpret **energy flow diagrams** to identify energy inputs, transformations, and outputs in a system.
- Identify energy forms (solar, chemical, electrical, thermal, mechanical, nuclear, gravitational) in real-world scenarios.
- Analyze diagrams of **power plants** to determine how energy is converted into electricity.
- Compare renewable and nonrenewable energy sources using charts, tables, and data sets.
- Interpret graphs showing **energy production, efficiency, cost, emissions, or resource use**.
- Use maps to identify the best locations for different energy technologies (solar, wind, hydroelectric, geothermal).
- Analyze case study information from major energy facilities to identify:
 - energy source used
 - energy transformations involved
 - advantages and limitations
 - environmental impacts

Scoring:

- Points will be awarded for each correct response.

Tiebreakers:

- Ties will be broken by the accuracy or quality of answers to select questions chosen by the event leader prior to the competition.

Resources:

- [Energy](#)
- [Physics for Kids - Energy](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles